Chapter - 1
Learning From Data

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1 Introduction

Let us look at the fundamentals of machine learning, utilizing an example from the domain of finance. Suppose a customer applies for a loan. Now, the bank must decide whether granting the loan is a good idea or not, since their goal is to maximize profit while minimizing risk. Naturally, the bank has no magic formula to predict whether a customer will make timely repayments or not. However, it is indeed possible to look at similar customers from the past and examine how their repayment behavior turned out to be.

The **key idea** here, is to use past data points to uncover an underlying pattern that connects customer information to their repayment behavior. So that, we can use it to predict the creditworthiness of fresh applicants. Using this example, let us now outline the main elements of the machine learning process :

1.1 The Underlying Pattern

As discussed above, in machine learning, the idea is to uncover the underlying pattern from a given set of data points. In the loan example, this underlying pattern is the relation that connects customer information with their repayment behavior. Although this relation is **unknown** to us, we do see glimpses of it through the available data points.

So, how do we describe this unknown pattern mathematically? Is it a deterministic function that always produces the same output for a given input, or is it something inherently probabilistic? To answer this, let us consider a simple practical question :

> **Can two loan applicants with identical information exhibit different repayment behaviours?**

Well, it is entirely possible. Two applicants may have identical ages, salaries and employment histories, yet one eventually repays the loan while the other defaults. This can happen due to factors that are not captured in our data, such as job loss, unexpected medical expenses or simply the inherent uncertainty in human life. This leads us to an important observation :

> **The scenario is probabilistic in nature.**

Therefore, we build our framework for machine learning accordingly, to capture that probabilistic nature from the very beginning. Instead of assuming a single correct output for every input, we describe the underlying pattern in terms of the probability of different possible outputs, for a given input. Mathematically, this is represented by a conditional probability distribution, formally called the **target distribution** :

$$P(y \mid \mathbf{x})$$

It is basically a function which gives us the probabilities of all possible outputs, for a given input.

Target Distribution Vs Target Function

To better understand target distribution, let us first consider a deterministic scenario. Suppose the bank follows a strict policy :

> **Approve every applicant whose annual income exceeds Rs. 7 lakh, otherwise reject the application.**

There is no uncertainty in this rule. Every input produces exactly one correct output. Such a relation can be represented simply, using a regular mathematical function. Let us call it the **target function** :

$$f(\mathbf{x})$$

For now, let us consider the annual income as the only input for simplicity. So, if x = Rs. 9 lakhs per annum, then :

$$f(9) = +1$$

Indicating that the loan is approved. However, we know real world decision making is rarely this simple. As discussed above, even if two applicants share the same data, their repayment behaviour may still vary because our data does not capture every single influencing factor. Therefore, instead of a fixed correct output, we get probabilities of different possible outputs, for example :

$$P(y = +1 \mid x = 9) = 0.80$$
$$P(y = -1 \mid x = 9) = 0.20$$

Indicating that an applicant with a salary of Rs. 9 lakhs per annum has an 80% chance of repaying the loan and a 20% chance of defaulting. Finally, it is important to **note** that the target function is simply a **special case** of the more general concept of target distribution. For example, when the probability of a certain output is 1 and thereby 0 for all others, the target distribution collapses to a target function :

$$P(y \mid \mathbf{x}) = \begin{cases} 1 & \text{if } y = f(\mathbf{x}) \\ 0 & \text{if } y \neq f(\mathbf{x}) \end{cases}$$

1.2 The Data Set

The data set is a collection of observations generated by the underlying probabilistic process. So, even though the underlying probability distribution is unknown to us, we do have examples generated from it in the form of our data set. In the loan approval example from above, each data point corresponds to a past customer and collecting N such data points gives us the required data set :

$$\mathbf{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), (\mathbf{x}_3, y_3)...(\mathbf{x}_N, y_N)\}$$

Where $\mathbf{x}_n$ is the application data of the nth customer, which is represented by an array of numbers as shown below :

$$\mathbf{x}_n = \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_d \end{bmatrix}$$

Here x_1, x_2 and so on upto x_d are the data attributes of a customer like age, salary, years in residence, outstanding debt and so on. While the corresponding output :

$$y_n \in \{+1, -1\}$$

represents whether that customer eventually turned out to be creditworthy or not. Now, the data set does not reveal the target distribution completely, but it provides us with examples using which we can approximate it. In other words, the data set acts as our limited window into the underlying pattern.

A coin toss does not reveal the coin's probability; a sequence of tosses provides evidence from which we can infer it. Likewise, a supervised dataset does not reveal P(y given x); it provides observed outcomes from which a learning model can infer some useful structure in it.

The dataset gives us empirical frequencies—how often particular observations occur in our finite sample. As the number of observations increases, those empirical frequencies can increasingly reflect the underlying probabilities.

You don't need repeated customers with identical entire profiles for learning to happen. So even though every customer may have a different x, the collection of many different examples gives the learning algorithm information about how the output tends to vary with the d input features.

2 Our First Learning Model : The Perceptron

So far, we have identified the two fundamental ingredients of the learning process, the **target distribution** and the **data set**. The next question is :

> **How do we use these observations from the data set to approximate the target distribution?**

As discussed above, Now, for a particular profile $\mathbf{x}_n$, the true underlying target distribution might be as follows :

$$P(y = +1 \mid \mathbf{x}_n) = 0.80$$
$$P(y = -1 \mid \mathbf{x}_n) = 0.20$$

The underlying phenomenon is probabilistic. Different learning models choose different ways of representing or approximating that phenomenon.

The target distribution is what we are trying to learn. With Perceptron, we are going to approximate it to a hard decision. We can also actually learn the probability distribution as we will see later on with other models.

The predictive rule doesn't have to reproduce all the information in the distribution. It's an approximation / decision mechanism. The hypothesis is our learned approximation or decision mechanism.

The perceptron doesn't care about modeling this underlying probability distribution. It just gets the final label $y = +1$ and that is what the perceptron will try to learn and optimize for. This is because perception is a simple linear model. Further there will be other advanced models that will try to capture the probabilistic aspect explicitly.

The perceptron does not attempt to estimate the target distribution itself. Instead, it learns a deterministic decision rule that predicts one label for every input. In other words, it compresses the uncertainty present in the target distribution into a single decision.

Linear classifiers such as the perceptron do not model the entire underlying target distribution. Instead, they optimize a linear decision boundary as per the labels assigned to the past data-points. It can be thought of as the perceptron approximating the target distribution, by learning the most probable label for a given x.

And the perceptron is deliberately not trying to recover the entire distribution. It is using those samples to learn a simpler deterministic decision rule.

2.1 Hypothesis Function

Now, what a perceptron basically does, is to assign a weight to each data attribute of the customer. These weights are multiplied to their respective data attributes and then summed up as follows :

$$w_1x_1 + w_2x_2 + ... + w_dx_d$$

The above linear summation of weights and attributes can be called the credit score of an individual customer. The idea is pretty simple, If the credit score is greater than a certain threshold, then we approve the loan, else we deny it :

$$w_1x_1 + w_2x_2 + ... + w_dx_d > \text{threshold (Approve Loan)}$$
$$w_1x_1 + w_2x_2 + ... + w_dx_d < \text{threshold (Deny Loan)}$$

This is equivalent to :

$$w_1x_1 + w_2x_2 + ... + w_dx_d - \text{threshold} > 0 \text{ (Approve Loan)}$$
$$w_1x_1 + w_2x_2 + ... + w_dx_d - \text{threshold} < 0 \text{ (Deny Loan)}$$

We can simplify the above expression further, by taking the minus of threshold as the zero[th] weight ($w_0 = -$threshold) and introducing an artificial attribute $x_0 = 1$ to go along with it, hence giving us :

$$w_0x_0 + w_1x_1 + w_2x_2 + ... + w_dx_d > 0 \text{ (Approve Loan)}$$
$$w_0x_0 + w_1x_1 + w_2x_2 + ... + w_dx_d < 0 \text{ (Deny Loan)}$$

Next, we re-write the above expression in vector (column matrix) notation to make it shorter and cleaner :

$$w_0x_0 + w_1x_1 + .. + w_dx_d = \begin{bmatrix} w_0 & w_1 & . & . & w_d \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ . \\ . \\ x_d \end{bmatrix} = \mathbf{w^T x}$$

The above formula outputs a real value that could be greater than, equal to or less than zero. Next, the above formula is wrapped inside a **sign** function. The sign is just a simple function that takes in a real number value and outputs $+1$ if the value is greater than zero and -1 if it is less than or equal to zero, hence giving us the final form of our perceptron hypothesis function :

$$\text{sign}(\mathbf{w^T x})$$

Hypothesis Set

2.2 Perceptron Learning Algorithm

Examining the perceptron hypothesis function, that we have arrived at above :

$$h(\mathbf{x}) = \text{sign}(\mathbf{w^T x})$$

We see that $\mathbf{x}$ representing the customer information is already a given, along with y, which is the correct loan decision in hindsight. So, the underlying pattern that we are trying to learn will be reflected in our choice of weights, as represented by the weight vector $\mathbf{w}$. Hence, our task now is to arrive at a set of weights, that are able to classify all given data-points correctly.

So, how do we compute the correct weights? First, we initialize the weights randomly, let us say initialize them all to zero. The perceptron learning algorithm then iterates through the dataset, looking for misclassified points $(\mathbf{x}_n, y_n)$, where :

$$\text{sign}(\mathbf{w^T x}_n) \neq y_n$$

Whenever such a misclassified point is found, the weight vector is updated by applying the following **update rule** :

$$\mathbf{w}_{new} = \mathbf{w} + y_n\mathbf{x}_n$$

This process continues, until no misclassified points remain in the dataset. And that is it, we then have our required set of final weight values.

The weights are therefore our learned representation of a useful pattern in the data

2.3 Update Rule Justification

Let us examine how the above update rule actually nudges the perceptron in the right direction at each misclassified point. If a point $(\mathbf{x}_n, y_n)$ is misclassified, it means :

$$\text{sign}(\mathbf{w}^\mathbf{T}\mathbf{x}_n) \neq y_n$$

Accordingly, the perceptron updates the weight vector as follows :

$$\mathbf{w}_{new} = \mathbf{w} + y_n\mathbf{x}_n$$

Substituting the updated weight vector from above and ignoring the sign function for now, we get the following :

$$\mathbf{w}_{new}^\mathbf{T}\mathbf{x}_n = (\mathbf{w} + y_n\mathbf{x}_n)^\mathbf{T}\mathbf{x}_n$$

Expanding the right-hand side, we get :

$$\mathbf{w}_{new}^\mathbf{T}\mathbf{x}_n = \mathbf{w}^\mathbf{T}\mathbf{x}_n + y_n\mathbf{x}_n^\mathbf{T}\mathbf{x}_n$$

This is equivalent to :

$$\mathbf{w}_{new}^\mathbf{T}\mathbf{x}_n = \mathbf{w}^\mathbf{T}\mathbf{x}_n + y_n||\mathbf{x}_n||^2$$

Therefore, for a misclassified point, where $y_n = +1$ and $\text{sign}(\mathbf{w}^\mathbf{T}\mathbf{x}_n) = -1$, it implies $\mathbf{w}^\mathbf{T}\mathbf{x}_n \leq 0$ and after the update we get :

$$\mathbf{w}_{new}^\mathbf{T}\mathbf{x}_n = \mathbf{w}^\mathbf{T}\mathbf{x}_n + ||\mathbf{x}_n||^2$$

The update is adding some positive value to $\mathbf{w}^\mathbf{T}\mathbf{x}_n$ which might make $\mathbf{w}^\mathbf{T}\mathbf{x}_n > 0$ as required. Although, it does not guarantee an immediate sign correction, it clearly nudges the classifier in the right direction.

Similarly, when $y_n = -1$ and $\text{sign}(\mathbf{w}^\mathbf{T}\mathbf{x}_n) = +1$, implying $\mathbf{w}^\mathbf{T}\mathbf{x}_n > 0$, we get :

$$\mathbf{w}_{new}^\mathbf{T}\mathbf{x}_n = \mathbf{w}^\mathbf{T}\mathbf{x}_n - ||\mathbf{x}_n||^2$$

Hence, showing that each update to the weight vector nudges the perceptron in the required direction, improving its performance on that particular data-point. The following image illustrates the above discussion :

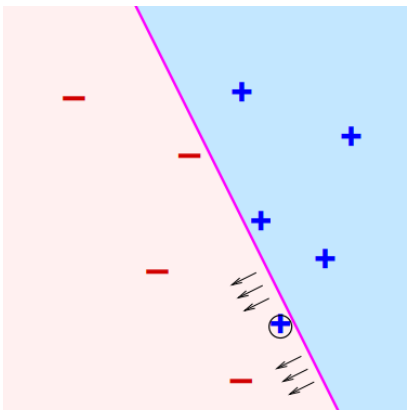


Figure 1: Perceptron

The pink line represents our linear classifier that is uniquely determined by our choice of weights. We have a misclassified **plus** point in the **minus** region and the update rule tries to correct this by nudging the classifier in the required direction.

Convergence Theorem

The convergence theorem for perceptron guarantees that if the data is **linearly separable**, repeated updates will eventually classify all points correctly.

Data being linearly separable simply means that, when the data points are plotted on a graph, it is possible to draw a straight-line that completely separates them into two distinct categories. The data in the above image for example, is linearly separable.

3 Types of Learning

As discussed in the introduction, machine learning is about learning patterns from data. Alternatively, we can put it as **using a set of observations to uncover an underlying process**. In either case, we have established that having a set of observations or data-points is a non-negotiable. However, the form in which the data is available to us can vary. Depending on this form and the nature of the learning task, machine learning is commonly divided into three main paradigms: supervised learning, unsupervised learning, and reinforcement learning.

3.1 Supervised Learning

In supervised learning, the model is trained on labeled data, where each example consists of : (**input, correct output**). The goal is to approximate a function from inputs to outputs that generalizes well to unseen data. For instance, the perceptron model above.

3.2 Unsupervised Learning

In unsupervised learning, the data is in the form of inputs only, without associated labels : (**input, no output**). The task is to discover hidden structures, patterns or clusters within the data. Examples include clustering, dimensionality reduction, and generative modeling.

3.3 Reinforcement Learning

In reinforcement learning, the model interacts with the environment by taking actions and receiving feedback in the form of rewards or penalties : (**action, feedback**). The model learns by trial and error to maximize its cumulative reward, gradually improving its decision-making. This setup is especially suited to sequential problems such as game playing, robotics and control systems.

4 References

1. CalTech Machine Learning Course - CS156, Lecture 1.

2. CalTech Machine Learning Course - CS156, Lecture 4.